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SAPE: Demystifying Sub-band Aware Power-Equalization for Cellular Networks

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ABSTRACT

Frequency-selective fading in LTE and 5G can cause severe Signal-to-Noise Ratio (SNR) variation across subbands, leading to higher Block Error Rate (BLER) and conservative Modulation and Coding Scheme (MCS) selection. While subband-level Channel Quality Indicator (CQI) enables finer link adaptation, current systems react to fading rather than proactively mitigating it. We implement a *Subband Power-Equalization at the transmitter based on subband-CQI feedback* in the srsRAN LTE stack, jointly pre-equalizing data and reference signals to flatten the channel response. Our design addresses quantized/delayed feedback and reference signal consistency, operating within existing protocol limits. The Software Defined Radio (SDR) testbed evaluation shows BLER reductions and throughput gains up to 25%, with the largest improvements in the worst 5–10% of channel realizations. These results demonstrate the practicality and effectiveness of proactive, PHY-layer frequency-selective compensation in real-time networks.

CCS CONCEPTS

• **Networks** → **Network resources allocation**; **Data path algorithms**; **Network experimentation**; **Network simulations**.

KEYWORDS

Subband Power Equalization, Channel Quality Indicator (CQI), Frequency-Selective Fading, Multipath Propagation, Software Defined Radio (SDR), srsRAN, Link Adaptation, 5G, LTE, Physical Layer Design

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1 INTRODUCTION

Modern LTE and 5G systems often suffer from severe multipath fading and frequency selectivity, especially in indoors or in urban environments, degrading reliability and throughput [11]. To mitigate this effect, LTE and 5G systems rely on frequency-domain equalization to compensate for channel-induced distortion and improve reliability in practical multipath environments [3]. Specifically, the use of Orthogonal Frequency Division Multiplexing (OFDM) transforms a frequency-selective channel into a set of parallel narrowband subchannels, each experiencing approximately flat fading. Then channel equalization approaches will be applied to each subchannel to reduce inter-symbol interference and enable robust demodulation even in severely frequency-selective channels. But the difference between subchannels brings challenges to the resource allocation and rate control.

There are two broad approaches to handling the frequency-selective channels. The first approach is treating all subchannels equally on the transmitter side, i.e., assigning the same amount of power and adopting the same modulation and coding. Then, the receiver will estimate the channel state information from reference signals and apply per-subcarrier frequency-domain equalization. This approach is simple, requires no feedback, and is widely adopted in most OFDM-based communication systems. However, the post-channel equalization techniques amplify noise along with the signal on the deeply faded subcarriers [8], degrading symbol reliability and leading to block decoding failures [11]. As shown in Fig. 1, these impairments constrain the system to select a conservative Modulation and Coding Scheme (MCS), thereby limiting spectral efficiency.

By contrast, classical waterfilling allocates more power to strong subcarriers and less to weak ones to maximize Shannon capacity, a result that presumes per-subcarrier rate adaptation so strong tones can use high MCS while weak ones use low MCS. However, LTE and 5G do not operated this way: a single MCS is applied to all Physical Resource Blocks (PRBs) scheduled for a User Equipment (UE) in a Transmission Time Interval (TTI), since wideband or subband Channel Quality Indicator (CQI) maps to one MCS choice for the entire Physical Downlink Shared Channel (PDSCH) allocation, not per subband [2]. Under this single-MCS regime, waterfilling is mismatched to the decoder: starving faded tones increases the likelihood that a codeword spanning all tones hits its decoding threshold, yielding higher Block Error Rate (BLER) and lower end-to-end throughput than uniform power allocation. Our measurements confirm this counterproductive effect in frequency-selective channels (Sec. 4.1.2).

In contrast, we propose a practical **Sub-band Aware Power Equalization (SAPE)** which proactively amplifies the deeply faded subcarriers before transmission. The design is based on subband-CQI feedback introduced by 3GPP since LTE Release 8 [2]. Using reported subband CQIs, we apply per-subband gain factors before OFDM modulation to pre-equalize deep fades. After going through the deep faded channel, the final signal will have more even distribution across subcarriers, which brings better performance, as shown in Fig. 1.

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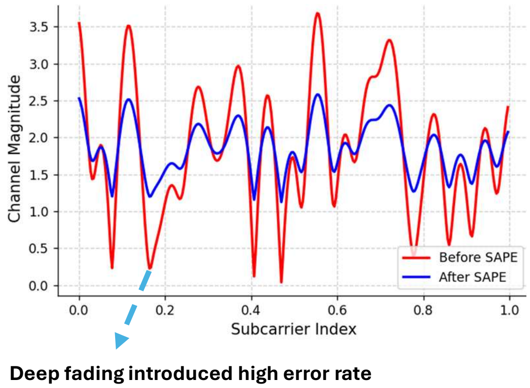


Figure 1: Channel magnitude response before and after applying the proposed SAPE in srsRAN.

To evaluate the effectiveness of the design and the compatibility of the existing 4G and 5G stack, we implement and evaluate with Matlab simulation on the Argos channel dataset and a real-world demonstration based on srsRAN using SDRs. The modification is only on the base station side, and there is no modification required on the UE side.

Based on the implementation, we expose the limitation of the traditional receiver-only equalization method under various Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) channel conditions and demonstrate up to 25% throughput improvement, especially in NLOS environments when augmented with our subband Power-Equalization technique.

We make the following contributions:

- We proposed the first pre-channel power equalization design under the cellular standard constraints.
- We implement the design through Matlab simulation and srsRAN to demonstrate the compatibility of the existing 4G and 5G stack.
- Our evaluation through real-world channel traces and real-world indoor environment shows that our approach outperforms the traditional receiver-only equalization and classical waterfilling approaches.

2 RELATED WORK

We compare and contrast our power equalization technique with the classic Waterfilling approach. Recent works such as [12] have explored self-optimizing water-filling algorithms tailored for hybrid fractional frequency reuse in 5G systems, combining theoretical capacity optimization with practical constraints. Other studies, for example [9], propose centralized water-filling–based power control schemes for two-tier cellular networks, allocating more power to users with favorable channel and interference conditions to maximize capacity. While these approaches are effective in theory, they rely on fine-grained power control, full channel state information, and advanced scheduling mechanisms, making them difficult to integrate into existing LTE/5G architectures. Waterfilling provides optimal power equalization only if each subcarrier is independently optimized for the best MCS to attain close to information-theoretic optimal performance. However, current LTE and 5G protocols don't allow sub-band aware MCS selection for a single user[1], limiting waterfilling to achieve its best performance [2].

In contrast, our approach leverages the sub-band CQI feedback opportunity reported in LTE Release 8 [2, 5] to extend the standard LTE framework to perform subband aware power equalization at the transmitter under the constraint of a fixed MCS per-user. Unlike classical waterfilling, which redistributes transmit power in a capacity-optimal manner, our method applies proportional adjustments based on reported CQI values. This adjustment modifies per-subband power while preserving proportional relationships, achieving optimal performance under the constraint of fixed MCS per subcarrier.

3 DESIGN

To effectively address the deep fading problem proactively, we propose to pre-equalize the transmit signal amplitude, i.e., preamplify the deep faded subcarriers so that there are no symbols that contain severe error information. To achieve this goal, we will utilize the subband CQI feedback, which has been supported since LTE.

3.1 Power-Equalization Algorithm

We provide an overview of our lightweight and real-time Power-Equalization algorithm which targets on filling the deep fading dips in the channel to improve the system performance of cellular system in the real world environments.

Per-subband Gain Normalization: Given the reported subband CQI for each subband i, we map it to a linear gain factor g_i using a configurable mapping function based on the SNR table taken from Table III in [7], specifically the column corresponding to Transmission Mode 111 with no retransmission (0 re-tx):

$$g_i = \min\left(10^{\frac{\mathrm{SNR}(\mathrm{CQI}i)}{20}},\ g_{\max}\right), \quad i = 1, 2, \ldots, N \tag{1}$$

Here, $g_{\max}$ is a predefined cap (e.g., 100) to prevent excessive amplification.

Gain Application to Data and Reference Symbols: The corresponding gain g_i is then applied to each PDSCH symbol $x_i[n]$ within subband i, i.e., $x'_i[n] = g_i \cdot x_i[n]$. Similarly, the same gain is applied to all reference symbols (CRS/DMRS) $r_i[m]$, i.e., $r'_i[m] = g_i \cdot r_i[m]$.

Total Power Normalization: To ensure that the total transmit power remains constant after applying different gain values to each subband, we compute a global normalization factor. This factor is derived based on the power before and after gain adjustment. Once calculated, it is applied uniformly to all subband signals — including both the modulated data symbols and reference signals — to scale them proportionally. This ensures that although the relative power across subbands changes, the overall power transmitted by the system stays the same.

This approach improves the uniformity of effective Signal-to-Interference-plus-Noise Ratio (SINR) across PRBs, aiding downstream MCS selection without changing the protocol or requiring precise channel knowledge. Crucially, it modifies modulated symbols directly rather than transmit power per subcarrier, operating entirely within the PHY layer and preserving Cell-specific Reference

[1]Subband aware MCS selection can be done across users

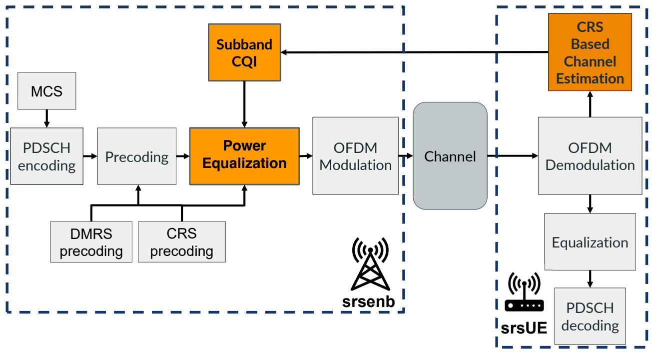


Figure 2: System diagram of the subband Power-Equalization mechanism integrated into the srsRAN 4G PHY stack. Orange block represents our modification in the srsran stack.

Signal (CRS)/Demodulation Reference Signal (DMRS) consistency via uniform scaling.

3.2 CQI feedback and Power-Equalization in srsRAN

To implement the proposed subband CQI-based Power Equalization scheme in a protocol-compliant manner, we modify the downlink processing chain in the srsRAN 4G stack [6]. The core modification involves extracting per-subband CQI values from the MAC layer scheduler and using them to apply amplitude-based shaping to the PDSCH signal in the physical layer.

As shown in Fig. 2, the process begins with subband SNR estimates at srsUE, which are mapped to corresponding subband CQI values. The selected subband CQIs, along with traditional wideband CQIs, are then fed back to the srsenb. In the original srsRAN pipeline, only the wideband CQI was fed back, which informs an MCS decision, which is held constant for a transmission burst.

We developed a pipeline to use the subband CQI information for Power-Equalization at srsenb. The Power-Equalization block is applied immediately after modulation and resource element mapping, but prior to Inverse Fast Fourier Transform (IFFT). It computes per-subband gain factors g_i based on the CQI values. These gains are then applied uniformly to both PDSCH data symbols and reference signals (including CRS, DMRS) within each subband to maintain consistency in channel estimation and equalization.

Unlike conventional receiver-side equalization which mitigates distortion after transmission, our method proactively applies gain shaping at the transmitter. Importantly, the Power-Equalization is compatible with HARQ, as the same gain profile is applied consistently to both initial transmissions and retransmissions of each transport block.

This approach leads to a flatter effective SINR distribution across PRBs, which in turn allows the system to more safely select higher MCS levels without violating reliability constraints. The design is lightweight, does not require protocol changes, and can be implemented entirely within the PHY layer of the open-source stack.

3.3 System challenges with srsRAN

In our implementation, we configure **4 subbands** for CQI reporting and Power-Equalization, balancing practical constraints with protocol compliance in srsRAN.

According to 3GPP specifications, subband CQI reporting is exclusively supported by PUCCH reporting types 1 and 1a, with a payload structure of $4 + L$ bits, where 4 bits encode the wideband CQI and $L = 2N_{\text{subband}}$ represents the differential subband CQI component [1].

In srsRAN, only a fixed-length implementation of reporting type 1 is supported for uplink CQI feedback. Under normal cyclic prefix (CP), the effective uplink control information (UCI) payload is constrained to 13 bits, limiting the differential component to 8 bits and thus restricting the number of supported subbands to four. This limitation is an implementation choice in srsRAN rather than a constraint imposed by the 3GPP specification.

Our 4-subband setup strikes a practical balance, enabling frequency-selective evaluation of Power-Equalization. As to be shown with the MATLAB simulation, a finer subband granularity supported by 5G can provide better performance. Future work may explore enabling extended PUCCH formats for finer subband granularity.

Furthermore, while 3GPP standards support subband CQI reporting [1], it is an optional feature that is often not implemented or enabled in commercial UEs. Most devices default to wideband CQI feedback due to lower complexity and signaling overhead. Similarly, subband CQI is disabled by default in many eNodeB configurations and can only be activated when UE capabilities permit. These constraints limit the applicability of subband-based mechanisms such as Power-Equalization in real deployments. Our evaluation, therefore, focuses on controlled testbeds like srsRAN where subband CQI reporting can be explicitly enabled.

4 IMPLEMENTATION AND EVALUATION

This section describes the practical system implementation using the srsRAN 4G stack, as well as complementary link-level simulations conducted using the MATLAB 5G Toolbox.

4.1 MATLAB 5G NR Simulation

We evaluate three configurations in our MATLAB-based link-level simulation: the original system, a subband power equalization scheme, and a water-filling-based power allocation. The simulations are conducted under fixed MCS settings (MCS 1–9), where the MCS table is adopted from prior work on SNR–CQI mapping [10]. Each experiment uses a fixed MCS for a given SNR point, without any link adaptation, to systematically explore the impact of different power control strategies. We use the publicly available Argos channel dataset from Rice University [4], which contains real-world downlink CSI traces collected from a massive Multiple Input Multiple Output (MIMO) testbed.

4.1.1 Simulation Configuration. We simulate a single UE in the downlink using the following configuration:

- Carrier: 52 PRBs (approx. 20 MHz), 30 kHz Subcarrier Spacing (SCS)
- Antennas: 4x4 MIMO (transmit × receive)
- Channel model: Argos channel dataset

- PDSCH: Type-A mapping, full-slot allocation
- Channel estimation: practical (non-perfect)

4.1.2 Simulation Results on MATLAB. Fig.4 compares the downlink throughput versus SNR comparison under different power control strategies, using our MATLAB-based simulation framework described earlier.

Notably, the power equalization method shows clear throughput improvement, especially under higher MCS levels. The gain is most evident for MCS 7–9, though here we only present MCS 9 as a representative high-MCS case. For MCS 3 and 6, modest gains are still observed.

In contrast, the water-filling approach, when applied without adjusting the MCS per subband, leads to performance degradation. In several SNR regimes, it performs significantly worse than the baseline system. This highlights a key limitation: without subband-aware MCS adaptation, aggressive power reallocation via water-filling may over-provision low-SINR subbands and under-power high-SINR ones, leading to higher block error rates and lower throughput.

4.2 srsRAN Testbed and Methodology

We prototype on the srsRAN 4G LTE stack with USRP B210 SDRs in a controlled lab loopback (Fig. 3). The system uses LTE Frequency Division Duplex (FDD) in Band 7, center frequencies 2680 MHz downlink and 2560 MHz uplink, 10 MHz bandwidth with 50 PRBs at 15 kHz SCS, and 1×1 Single Input Single Output (SISO). Frequency selectivity is summarized by one wideband CQI and four evenly spaced subband differential CQIs across the 50 PRBs. The eNB employs time-domain round-robin scheduling, and CQIs are reported periodically (every **40 ms** by default in srsRAN).

Each trial transmits downlink User Datagram Protocol (UDP) traffic using `iperf` with a single, fixed MCS chosen from the reported CQIs. We compare a baseline with receiver-side equalization only against our subband Power-Equalization, which applies per-subband gains uniformly to PDSCH and CRS/DMRS prior to the IFFT. The eNB logs BLER and application-layer throughput. If both modes yield indistinguishable performance, the MCS is increased and the run is repeated to avoid overly conservative operation. We study two representative channels, an indoor LOS link with minimal obstruction and an indoor NLOS setting with deliberate blockers at two placements (POS1, POS2), to test the hypothesis that Power-Equalization is neutral in flat, high-SNR conditions ((as shown in Fig. 3c) yet beneficial under frequency-selective multipath(as shown in Fig. 3a and Fig. 3b).

The downlink scheduling is time-domain round-robin. CQI reporting is enabled in periodic mode with one wideband CQI and 4 subband differential CQIs. All experiments are conducted in a controlled lab environment using a loopback setup with USRP B210, allowing for deterministic evaluation of the Power-Equalization mechanism.

4.2.1 Implementation Gap in 5G SDR Systems. Although the proposed Power-Equalization technique is applicable to both LTE and NR, practical over-the-air deployment in 5G remains infeasible. The challenge lies not in the algorithm itself, but in the limitations of current open-source SDR platforms. **Popular frameworks such as srsRAN 5G and OpenAirInterface (OAI) do not yet support periodic subband CQI reporting or the necessary UCI encoding over PUSCH/PUCCH.**

While 3GPP NR specifications (e.g., TS 38.212, TS 38.213) define these feedback mechanisms, their lack of implementation in public toolchains hinders prototyping. **We therefore choose the mature srsRAN 4G stack for real-world evaluation, which fully supports periodic CQI feedback with subband granularity.**

4.3 Experimental Results in srsRAN

4.3.1 Indoor NLOS (Heavy Multipath). This scenario evaluates our system in severe multipath conditions where the UE is completely blocked from the eNB's direct line-of-sight. Two different positions were tested in this environment.

Position 1 (POS1). The results below correspond to Position 1 (POS1) as seen in Fig. 3b.

The aggregated summaries in Fig. 5a and Fig. 5b illustrate the effect of subband-based Power-Equalization across different MCS levels. In Fig. 5a, we observe that for MCS 11 and MCS 15, Power-Equalization consistently reduces the median BLER and suppresses high-error outliers, suggesting a clear reliability gain. However, at MCS 13, the performance gap between the two configurations is minimal. While the median BLER remains comparable, the distribution with Power-Equalization exhibits slightly higher spread. This is partly due to a few high-BLER samples identified as outliers in the boxplot, which amplify the perceived variability. These outliers may stem from measurement noise or specific fading conditions where the gain adjustment has limited effect.

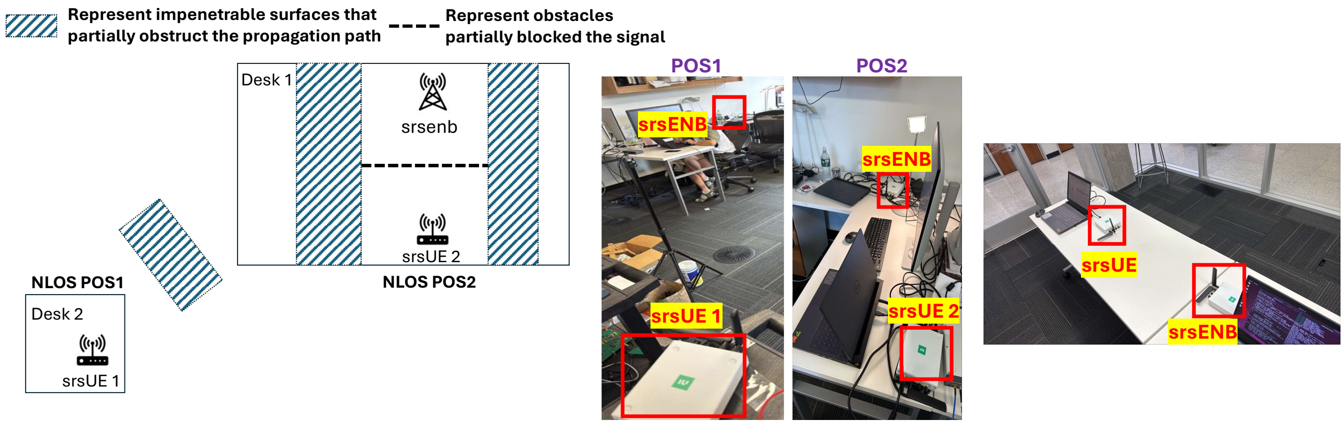
Similarly, the throughput distributions in Fig. 5b confirm this trend. For MCS 11 and MCS 15, Power-Equalization improves the median throughput and significantly reduces the spread of low-throughput samples. In contrast, MCS 13 exhibits comparable performance in both settings, showing that subband gain adaptation offers the most benefit when the MCS is more aggressive and susceptible to multipath-induced degradation.

Overall, these results suggest that Power-Equalization is particularly effective in mitigating channel selectivity at higher spectral efficiency settings, while its impact is less pronounced when the baseline MCS is already conservative.

Together, these results highlight that subband Power Equalization not only improves average performance but is particularly effective in enhancing tail-end reliability and throughput under heavy multipath conditions.

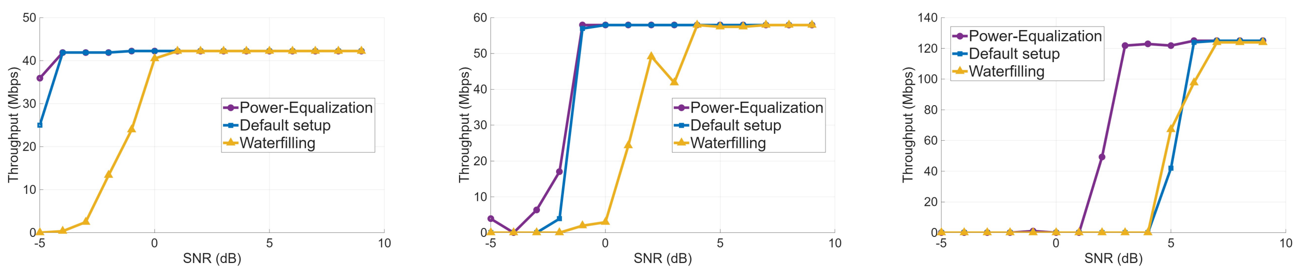
Position 2 (POS2). The results below correspond to Position 2 (POS2) as seen in Fig. 3b.

Fig.6a and Fig.6b show the block error rate and throughput Cumulative Distribution Function (CDF) plots at MCS 9. The CDF of throughput exhibits a clear improvement in the lower percentiles: at the 5$^{\text{th}}$ and 10$^{\text{th}}$ percentiles, the throughput with Power-Equalization is noticeably higher than that without it. This indicates that the worst-performing samples, where the channel conditions are most adverse, benefit the most from subband-based Power-Equalization.



(a) Floor plan of the indoor NLOS test environment. (b) Indoor NLOS. (c) Indoor LOS (Clean).

Figure 3: Experiment setup with srsRAN LTE stack and USRP SDRs in various multipath environments.



(a) MCS 3 (b) MCS 6 (c) MCS 9

Figure 4: Throughput vs. SNR comparison for MCS 3, 6, and 9 under different power control strategies.

Similarly, the CDF of BLER confirms this trend: in the 90th to 95th percentile region, BLER with Power-Equalization is significantly reduced, highlighting enhanced reliability in the worst-case transmission instances.

Overall, this experiment demonstrates that in highly reflective, NLOS environments, subband Power-Equalization provides clear gains in both throughput and reliability—particularly in the tail end of the distribution where robustness matters most.

4.3.2 ***Indoor LOS (Clean).*** We used a pristine LOS link with minimal multipath (Fig. 3c) and stressed the downlink at MCS 28 to test the upper performance bound. As hypothesized for a flat, high-SNR channel, throughput CDFs for baseline and subband Power-Equalization are virtually indistinguishable (Fig. 7a), and BLER curves show only a slight reduction in rare error bursts with Power-Equalization (Fig. 7b).

Conclusion. When frequency selectivity is negligible, subband Power-Equalization provides no throughput gain, yet it does not degrade performance and yields marginal robustness in edge cases. This establishes a no-regret behavior in benign channels, while motivating its use primarily in frequency-selective conditions.

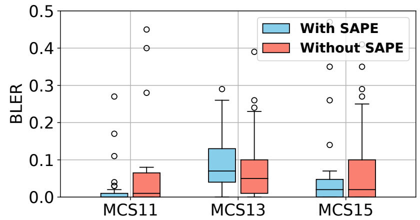
These results suggest that subband Power-Equalization is most beneficial in environments with severe frequency selectivity and high spatial multipath richness, where different frequency components experience vastly different fading conditions. In such cases, the ability to dynamically reshape power across subbands can significantly enhance worst-case performance, boosting robustness and user experience for the weakest links in the system.

5 CONCLUSION AND FUTURE WORK

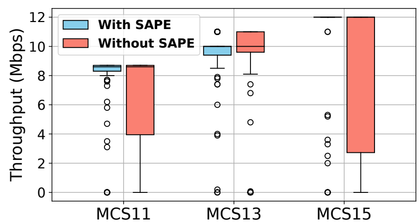
This study provides a practical, lightweight, and real-time power equalization strategy built into the srsRAN software stack, offering useful insights into system-level challenges, 3GPP compliance, and real-world evaluation. There are several promising directions for future work.

First, while this work focuses on transmitter-side power equalization, we do not explicitly perform subband-aware MCS adaptation. Our results suggest that, under the same channel conditions or SNR levels, power equalization enables the system to operate reliably at higher MCS levels, leading to improved throughput and reduced BLER in multipath-rich environments. A natural next step is to jointly design subband-aware power equalization and MCS adaptation to further improve link efficiency.

Second, this work evaluates power equalization in srsRAN with a single-user SISO setting. Importantly, **SAPE operates independently of the scheduling layer and does not alter existing**

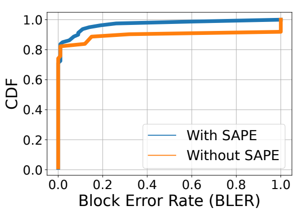


(a) BLER across MCS levels with and without SAPE.

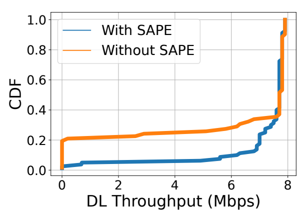


(b) Throughput performance across MCS levels with and without SAPE.

Figure 5: Indoor NLOS (Heavy Multipath) - POS1 Results.

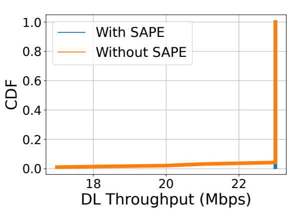


(a) BLER CDF for MCS 9

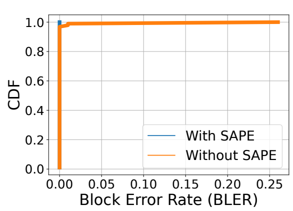


(b) Throughput CDF for MCS 9

Figure 6: Indoor NLOS (Heavy Multipath) - POS2 Results.



(a) Throughput CDF for MCS 28



(b) BLER CDF for MCS28

Figure 7: Indoor LOS (Clean) Results.

resource allocation or scheduler behavior. Extending the proposed design to multi-user MIMO (MU-MIMO) scenarios presents additional challenges, including inter-user interference management, scheduler interactions, and fairness-aware resource allocation. Understanding how subband-aware power equalization interacts with MU-MIMO precoding and scheduling policies is non-trivial and remains an important direction for future system-level studies.

Finally, we plan to explore the impact of subband granularity through extensive simulations. Different subband sizes introduce trade-offs between frequency selectivity, feedback overhead, and computational complexity. A systematic evaluation of subband granularity can provide practical guidelines for selecting appropriate configurations under diverse channel conditions and deployment scenarios.

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